



Where does AI play a major role in the new product development and product management process?

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Abstract

The integration of Artificial Intelligence (AI) into various industries and domains is a pertinent and contemporary topic extensively discussed in the scientific literature. One such domain is product management, particularly focusing on New Product Development (NPD). The intersection of AI and product management presents substantial opportunities for research into the application of AI solutions to support product management activities. Drawing on a structured literature review of 190 publications and interviews with five experts in AI and product management, we map out AI applications across different phases of the new product development process. The findings indicate that significant results can be achieved only by disaggregating AI and product management into specific methods and phases and reveal a predominance of AI methods such as sentiment analysis, knowledge extraction, and demand forecasting in early-phase activities, with fewer studies examining AI's application in later stages such as product testing, validation, and post-launch optimization. The study identifies several research gaps, including the underutilization of AI in concept testing, the limited focus on integrative AI solutions that span multiple NPD phases, and the absence of systematic frameworks for AI-driven product management. We propose a set of guidelines to facilitate the development of a cohesive, end-to-end framework. By situating AI within a broader managerial and organizational context, this paper offers both theoretical insights into AI's evolving role in product development and practical recommendations for stakeholders seeking to leverage AI's transformative potential.

Keywords Artificial Intelligence · Machine learning · Product management · AI-driven management · New product development · Literature review

JEL Classification O31 · O32 · O33 · M15 · M30

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1 Introduction

Over the past decade, the burgeoning interest in Artificial Intelligence (AI) has signaled not merely a technological shift but also a fundamental reorientation in how organizations conceive product development and success. The surge in AI-focused publications (Yoon et al. 2022; Prahani et al. 2023; Prieto-Gutierrez et al. 2023) attests to a widespread belief that AI can unlock novel sources of both tangible and intangible value in product management. Yet, the assumption that AI seamlessly integrates into existing processes invites deeper scrutiny, particularly from a managerial perspective. While intangible benefits—such as cultural shifts or competitive advantage—can reshape an organization’s innovation climate, they also place new demands on managers regarding stakeholder engagement, team skill-building, and ethical oversight. On the tangible side, managers often seek cost reduction, revenue growth, and customer satisfaction improvements, but the infrastructure, continuous data updates, and specialized capabilities required for AI’s effective deployment underscore how success metrics in product management may need to be recalibrated.

From a managerial standpoint, AI’s transformative potential also raises questions about the alignment between traditional product development frameworks and the emerging data- and analytics-driven practices. Although long-standing success factors—strategic vision, robust new product development (NPD) processes, and skilled human capital—remain crucial (Soltani-Fesaghandis and Pooya 2018), AI amplifies the need for cross-functional collaboration, governance structures, and iterative learning mechanisms (McKinsey 2021). Managers may find that the integration of AI reshapes not only their operational workflows but also the broader organizational mindset around risk, agility, and customer centricity.

Scholarly inquiry into AI’s role in product management has thus far been split between research on managing AI-based products (Frank et al. 2021) and on using AI to enhance product development itself (Füller et al. 2022; Truong and Papagianidis 2022). While the former has gained traction, the latter remains underexplored, leaving open crucial questions about how AI might alter core managerial practices—from idea generation and concept testing to commercialization and lifecycle management, as there is a dearth of studies that have examined the topic systematically, or that have aimed to map existing knowledge and identify new avenues for research. In response, this article explores how AI solutions can be woven into the product development and management process, with a particular focus on identifying research gaps and outlining what it would take to establish a unified AI-driven framework. Specifically, we address two research questions (RQ):

RQ1 What are the current major themes and research gaps in the literature when it comes to AI that helps with product development and management processes?

RQ2 What would it take to create a common AI-driven product development and management framework?

By synthesizing diverse perspectives, this study aims to inform both scholars and practitioners on the evolving interface between AI technologies and product development and management. The structure of the paper is as follows: Sect. 2 lays out definitions of AI and product management concepts, breaking it down into specific methods and phases. Section 3 describes the research methodology, Sect. 4 maps existing AI applications across product development and management stages, and Sect. 5 provides a critical discussion with theoretical and managerial implications, limitations, and agenda for further research. Finally, Sect. 6 summarizes the paper.

2 Theoretical background

In the scientific literature, the phrase "AI-driven product management" or "AI-driven product development" is not readily identifiable, as the terminology is overly broad. The term "AI-driven" is often misinterpreted or conflated with "data-driven" in the literature, leading to confusion between the application of AI techniques in a specific business domain and the use of data in that domain. To gain insight into the AI-driven product development and management concept, it is necessary to decompose the definitions of Artificial Intelligence and product management into their constituent solutions and phases.

2.1 Artificial Intelligence

While Artificial Intelligence (AI) can provide many benefits to the organizations, it is not an easy technology to use or implement compared to other digital technologies. To create and improve new products, organizations must manage vast amounts of data and possess the appropriate infrastructure for this task (Bosch et al. 2018). Based on interviews conducted by Figalíst et al. (2020), only 9% of respondents indicated that the effort required to establish the infrastructure for software analytics and AI-based business intelligence is relatively low or very low. In contrast, over 50% reported that it is high (37%) or very high (14%). The attempt to establish the proper infrastructure presents organizations with challenges not only from a financial perspective but also due to technical complexity (Figalíst et al. 2020). There are several other challenges associated with implementing AI solutions, such as the quantity and quality of data (Omraní et al. 2022), trust and security, human factors (Woschke and Haase 2016), and the lack of adequate infrastructure or skills in the team (Erebağ and Turgut 2021).

Artificial intelligence has made significant strides in various sectors, e.g., AI is being increasingly used in healthcare for disease diagnostics, hospital record management, living assistance, and biomedical research (Chatterjee et al. 2022). AI also has the potential to revolutionize the agri-food sector through precision agriculture, crop monitoring, predictive analytics, supply chain optimization, food processing, and personalized nutrition (Taneja et al. 2023) or manufacturing sector by improving processes, reducing costs, and making operations more efficient (Vaia et al. 2021). AI plays a crucial role in companies of different sizes,

including small and medium-sized enterprises (SMEs), start-ups, and large companies. SMEs recognize the potential of AI but face challenges such as lack of knowledge, costs, and inadequate infrastructure (Borah et al. 2022; Oldemeyer et al. 2024), while start-ups can estimate their potential for AI and apply AI procedures with a minimum of effort using AI tools (Schuh et al. 2019). Large enterprises exhibit a higher level of AI adoption across various dimensions, including AI adopting strategies and utilizing AI technologies for project management purposes (Tominc et al. 2024). Challenges arise both before implementation and during its operation (technical and social), requiring all types and sizes of organizations to be prepared by developing AI maturity (or readiness) (Lokuge et al. 2019; Jöhnk et al. 2021; Witkowski and Wodecki 2024). Maturity and readiness assessment tools can help organizations identify gaps in their AI capabilities, accelerate the implementation process, and ensure they are progressing towards their AI-related goals (Molla et al. 2009).

There are many definitions of Artificial Intelligence, but from the perspective of this article, what is essential is the use of AI as an aid in the product management process. Hence a definition is needed that breaks down the concept of Artificial Intelligence into several smaller fields. Bernijazov et al. (2021) note that between 2010 and 2020, there was a clear shift towards using data-centric methods, such as machine learning or deep learning, rather than traditional knowledge-based methods, due to the increased availability of data and increased computing power. This has resulted in machine learning or deep learning solutions being widely used in other fields, for example, the YOLO algorithm (Redmon et al. 2016) used for object detection is based on deep learning's convolutional networks and could be considered a computer vision solution. This shift prompted to seek a more recent division of AI. Zhang, et al. (2021a) conducted interviews and identified six commonly used types of Artificial Intelligence in organizations during the New Product Development (NPD) process:

- Analytic AI (demand forecasting, risk assessment, and sentiment analysis),
- Functional AI (Internet of Things solutions and robots),
- Interactive AI (smart personal assistants and chatbots),
- Text AI (speech-to-text conversion, text recognition, and machine translation),
- Visual AI (computer vision and augmented reality), and
- Robotic AI (robotic process automation).

This approach, despite being significant, employs generic names in some cases from the perspective of scientific literature search engines (e.g., "Functional AI" is challenging to equate with Internet of Things solutions), and it can be observed that it blends the usage of AI as a support in the NPD process with managing the products that possess AI components. Artificial Intelligence is an extensive field and is used in various domains with distinct meanings. As previously noted, from the perspective of this research, establishing this division is imperative. Thus, based on the theory presented, this division is discussed in Sect. 3 on the research design.

2.2 Product management and new product development

According to Kotler (2006), a product is "anything that can be offered to a market for attention, acquisition, use or consumption that might satisfy a want or need. It includes physical objects, services, persons, places, organizations, and ideas". However, the term product is commonly understood as a physical or virtual object, or a combination of both, such as a smart, connected product (Porter and Heppelmann 2015). In the research presented in this article, the definition of a product is not limited to any specific type.

According to Hughes & Chafin (1996) "derived from strategic product planning, initiation and market introduction of product innovations, alongside with maintenance and improvement of existing products, are the central tasks of product management". If we were to consider the literature, however, biggest part of research on product management primarily focuses on New Product Development (NPD) activities, which can be defined as the process of bringing new products to market (Elif Karakoylu et al. 2020). One of the most popular models for new product development is the stage-gate (or phase-gate) model, created by Robert Cooper, used successfully since the 1980s. It assumes that the process is divided into phases, in which work is done, and gates or decision points between these phases, in which the quality of the previous phase is checked and a decision is made on whether the next phase should be started (Rao et al. 1999). However, the exact phases and their division can vary in the scientific literature. There are many different divisions in the scientific literature. R. Cooper (1998) notes that "The new product process is flexible — where stages and decision points can be skipped or combined, as dictated by the nature and risk of the project". Some authors are referring that the product must be primarily designed (product design phase) and created (product implementation phase), and all activities that happen at the beginning and end can be referred to as the "fuzzy front end" (Reid and de Brentani 2004), sometimes referred to as a "fuzzy front end of innovation"; or the "fuzzy back end" (product commercialization phase). The different approaches to process partitioning are presented in Table 1.

The NPD process consists of several phases, which have undergone changes and variations over time. The initial phases of the NPD process include idea generation and screening, as well as internal strategy and assessment. The idea generation phase, also known as ideation (Cooper and Edgett 2008), idea management, idea sourcing, product discovery, idea development, or concept generation, involves a systematic search for new product ideas both internally and through sources such as customers, competitors, or suppliers (Kotler 2006; Cooper and Edgett 2008). Once a number of ideas have been generated, the idea screening phase reduces them to the best options. Some literature sources consider this a phase of the NPD process (Kotler 2006; Trott 2017; Ozcan et al. 2021), and others as a gate between phases (Cooper 2010). Some sources indicate that companies should have internal strategies regarding new product development before all idea generation activities. Part of this strategy should be continuous market research and investigation of market dynamics (Annacchino 2003).

With the best ideas collected, companies build a business case—they prepare a product description or definition, sometimes also called the product protocol, and

Table 1 Different approaches to NPД across decades (*commerc.* = *commercialization*)

	Noori et al. (1997)	Griffin (1997)	Annacchino (2003)	Kotler (2006)	R. G. Cooper (2010)	Crawford and Di Benedetto (2010)	Trott (2017)	Zhang et al. (2021a)
Phase 1	Strategic planning	Idea/concept generation	Market assess- ment and inter- nal assessment	New Product Strategy	Discovery	Opportunity identification and selection	Idea generation	Idea development
Phase 2	Opportunity identification	Idea screening	Prototyping	Idea generation	Scoping	Concept Genera- tion	Idea screening	Business analysis
Phase 3	Opportunity development	Business analysis	Core development and testing	Idea screening	Build the busi- ness case	Concept/project evaluation	Concept testing	Product design
Phase 4	Product/service creation	Development	Industrialization	Concept develop- ment and testing	Development	Development	Business analysis	Product testing
Phase 5	Product/service introduction	Test and valida- tion	Commerc	Marketing strat- egy	Testing and vali- dation	Launch	Product develop- ment	Commerc
Phase 6	Life cycle man- agement	Commerc	Optimization	Business analysis	Launch		Test marketing	Post–commerc
Phase 7				Product develop- ment			Commerc	Operations man- agement
Phase 8				Test marketing			Monitoring and evaluation	
Phase 9				Commerc				

evaluate the sales opportunities of the new product, costs, and potential benefits to see if the creation of the new product is profitable and aligned with the company's goals. Some articles also indicate concept testing or concept evaluation in this phase (Cooper 2010; Crawford and Di Benedetto 2010), presenting a product concept to a group of target customers. A simple description of the solution is often enough to conduct such a test, but the more precise and specific the presentation of the concept to customers, the greater the confidence that the product will be successful.

After the first phases, which, as previously mentioned, can be referred to as the “fuzzy front end”, the next phase is the design and creation of the product, that is, the conversion of a concept into a working product with predictable performance. In this phase, a product prototype or MVP, product specification, implementation plans, and sometimes even marketing plans are created (Crawford and Di Benedetto 2010). The next phase is the testing and validation phase—the product, marketing program, and sales operations are tested (Cooper 2010) in actual market realities. Sales trials may also be conducted in this phase, and positioning strategy, product advertising, and price positioning are checked. Organizations check what the first reactions of customers will be and how they will use the product. All this information can lead to better sales and profit forecasting. In cases where product development costs are low or where management is confident in the product's success, the length of this phase is reduced to a minimum, or the testing phase is omitted altogether (Kotler 2006).

In many NPD models and approaches, the commercialization phase is considered the final stage of the NPD process, where the product is introduced into full-scale production, marketing, and sales. Organizations often face several key questions during the launch of a product, including the timing of the launch, the geographic scope of the launch, target customers, and the accompanying activities and actions. After the product has been introduced, organizations monitor its performance in the market and may make necessary improvements. Additionally, the NPD process and related operations may be optimized. While some product researchers include this phase as the last phase of NPD (Noori et al. 1997; Annacchino 2003; Trott 2017; Zhang et al. 2021a), others consider it a part of product lifecycle management (PLM). Considering all activities from all phases, it was decided to include a fairly large number of activities from the area of product management while simultaneously attempting to narrow the search to ensure it was not too general from the perspective of scientific article search engines. This approach also aimed to avoid topics that could be studied separately as large scientific fields (e.g., sales or marketing).

Approaches to the NPD process vary, but as previously mentioned, all focus on the fuzzy front end, encompassing initial phases related to idea generation, business analysis, and concept testing, followed by product design and creation, and then the fuzzy back end, which typically involves product testing and implementation. Product Lifecycle Management (PLM) can be approached in numerous ways. Generally, PLM can be defined as the business activity of managing a company's products in the most effective way across all stages of their life cycles (Stark 2016). The first product life cycle model, inspired by biology, emerged in the 1960s and divides it into four phases (Kotler 2006; Cao and Folan 2012):

- Introduction—a phase of slow sales growth with no profits due to high costs of launching a new product into the market;
- Growth—a stage of rapid market acceptance and increasing profits;
- Maturity—a stage of sales slowdown as the product has reached all potential customers, with reduced profits due to high marketing costs to defend the product against competition; and
- Decline—a phase of withdrawal, where sales and profits decline.

In contemporary literature, especially in so-called grey literature or among organizational white papers, PLM is understood quite differently—as computer tools focused on the design and broad management of products (Cao and Folan 2012). From this perspective, the history of PLM dates back to the 1970s and 1980s with the development of computer-aided tools supporting product design—Computer Aided Design (CAD), Computer Aided Manufacturing (CAM), and Computer Aided Engineering (CAE), along with the development of Product Data Management (PDM) systems to control and manage data and information related to products. Over time, these tools were joined by Enterprise Resource Planning (ERP), Customer Relationships Management (CRM), and Supply Chain Management (SCM) systems, each focused on a specific process in the product life cycle (Ameri and Dutta 2005; Cao and Folan 2012). Integrating these disparate systems and tools into one PLM system has become the specialty of current manufacturers, which also influences ongoing research in this area. In this understanding of PLM, literature includes articles on how to utilize PLM in the NPD process to achieve success in a given industry (Vezzetti et al. 2017; Tai 2017; Pinna et al. 2018). Due to this understanding of PLM in contemporary literature, the terminological confusion, and the previously mentioned disarray concerning the division, the term PLM is not considered in the continuation of this article. Considering this fact, during the literature review in the next chapter, the keywords *product management* are examined, which is a broader topic from the perspective of scientific article search engines and encompasses the keywords *product lifecycle management*.

3 Research design

The research design for this paper combines a literature review with expert insights to explore the current landscape of AI applications in product development and management. Apart from reviewing the existing academic literature, we also incorporated expert opinions to enrich our understanding of the field. For the study to achieve its goals, i.e., to provide researchers and practitioners with a structured, categorized view of what has been discovered in the AI-driven product development and management literature, there are two proposed research questions. The first one is formulated as follows:

RQ1 What are the current major themes and research gaps in the literature when it comes to AI that helps with product development and management processes?

As the analysis of the literature provides results related to different use cases of AI in different phases of the product development and management process, wanting to structure them, a discussion related to how to do it follows. Thus, the second question has the following formulation:

RQ2 What would it take to create a common AI-driven product development and management framework?

In addition to analyzing the existing literature, five experts with experience in both AI deployment and product management were consulted to deepen and validate the insights derived from the review. Each expert was asked two open-ended questions:

1. *What are the current major themes when it comes to AI that helps with product development or management?*
2. *What would you think future research should focus on, when it comes to AI that helps with product development and management?*

Their qualitative responses were integrated with the literature-based findings to enrich the discussion and highlight practitioner-relevant priorities.

The selection of keywords connected with product development and management is based on the definitions in the previous section derived from the literature sources including product development phases and product management to expand search results. In order to decide on the keywords connected with AI, we firstly decided to use the *VOSViewer* tool, which shows the links between keywords from given articles. To do this, we searched the *Scopus* database by entering a query (*artificial AND intelligence AND ((product AND development) OR (product AND management))*) and found 6867 articles, which we imported into the *VOSViewer* tool—the resulting map is shown in Fig. 1. While the map shows more of the areas in which these terms together are most commonly used (agriculture, economics, manufacturing, medicine), it also confirms initial assumptions about certain keywords (e.g., prediction or expert systems).

The literature review consists of a structured search and identification process based on Brocke et al. (2009) and Webster & Watson (2002). We searched for articles using the *Scopus* database, which we chose because of its repeated use in other similar articles focused on the use of AI in business (Borges et al. 2021). Based on the earlier linkage map and considering previously discussed research in this area (Zhang et al. 2021a), we decided to use two divisions for our research—into general subfields of AI usage and specific methods listed below. General subfields of AI used are *Text* (Natural Language Processing area), *Vision* (Computer Vision area), *Prediction*, *Optimization* and *Automation*. Specific methods and use cases used are *demand forecasting*; *anomaly detection*; *sentiment analysis*; *recommendation systems*; *image recognition*, *object identification*, and *object tracking*; *knowledge extraction*; *conversational AI*; *autonomous systems and robots*. However, as some of the keywords are vague (e.g., ‘optimization’, ‘text’), this results in a vast number

of results, hence we adjusted them accordingly. The keywords used in the literature review are summarized in Table 2. We also considered many other methods, such as topic modelling, but dropped it due to the fact that it appears with methods such as sentiment analysis, and in.

The following inclusion criteria were applied: (i) a focus on AI methods and techniques used in the new product development (NPD) process, (ii) restriction of the publication timeframe to the last 8 years (2017–2025) to incorporate the most recent research, (iii) selection of studies within the field of management sciences (categorized in search engine under Business & Management, Social Sciences, and Decision Sciences), and (iv) limitation to publications written in English. Exclusion criteria were also defined, which included: (i) lack of access to the full text of the article, (ii) "false positive" publications (e.g., cases where the acronym AI referred to something other than the artificial intelligence), and (iii) articles unrelated to products or product development in any way. To ensure the quality of the selected

Table 2 Keywords from the fields of AI and Product Management used in the literature review process (* means either s or z)

Field	Keywords
Product Management	Idea generation, ideation, idea management, idea sourcing, product discovery, idea development, concept generation Concept testing, concept evaluation, business analysis Product design, product development Product testing, product validation Product launch, product commerciali*ation Product optimi*ation, product improvement, product operations management New Product Development, NPD Product Management
AI	Predictive analytics, prediction Natural Language Processing, NLP Computer Vision Automation AI optimi*ation, Artificial Intelligence optimi*ation Machine Learning, Deep Learning, Supervised Learning, Unsupervised Learning, Reinforcement Learning Recommendation system, recommendation engine Demand forecasting Image recognition, object identification, object tracking Sentiment analysis Knowledge extraction, information extraction Conversational AI, chatbots, expert systems, questions answering Autonomous system, robotics Anomaly detection

articles, only publications from peer-reviewed academic journals or conference proceedings were considered. The selection criteria were first applied to titles and abstracts during the initial screening phase, followed by a full-text evaluation in the qualification stage. The initial keyword-based search yielded a total of 1533 results (as of February 6, 2025). After removing duplicates, analyzing titles and abstracts, and excluding publications that did not meet the specified criteria, 204 articles were shortlisted for full-text review. Subsequently, after a more detailed verification process, 48 research papers were excluded, leaving a final selection of 156 publications. The most common reason for exclusion was lack of direct relevance to AI methods usage in the context of new product development. Next, an analysis of cited sources in the selected publications and publications that cited the identified studies was conducted (backward and forward search), leading to the identification of 25 additional relevant articles. Nine more relevant publications were identified through an additional search using Google Scholar with selected keywords. As a result of this selection process, a final database of 190 articles was compiled. A graphical representation of the literature review process is presented in Fig. 2.

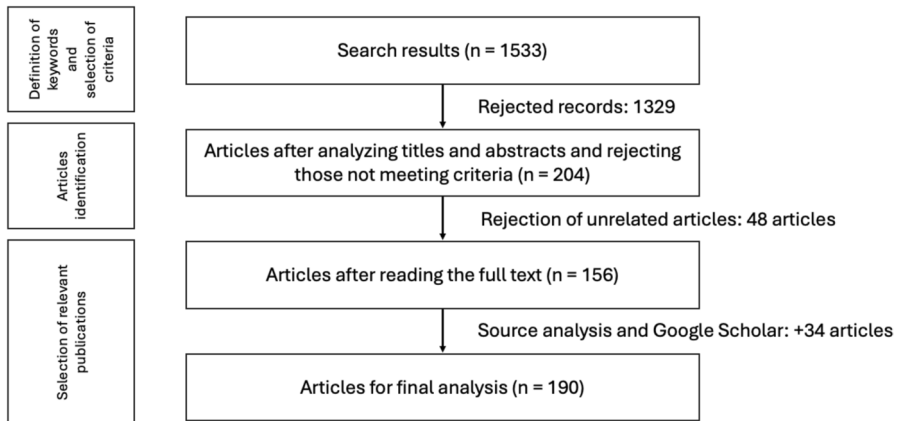


Fig. 2 Literature review process based on Brocke et al. (2009) and Webster & Watson (2002)

4 Results

This section presents the combined findings from both the literature review and the expert interviews. First, it synthesizes the insights gained from analyzing specific subfields and use cases of AI in various phases of product development and management. Second, it incorporates the perspectives of the five interviewed experts, who provided qualitative reflections on current major themes and future directions for AI-driven product management. We began by categorizing AI methods according to their typical use cases (e.g., demand forecasting, sentiment analysis, knowledge extraction) and then mapped these to corresponding subfields (e.g., Natural Language Processing, Computer Vision). For instance, in the “text” subfield related to NLP, no additional methods emerged beyond the already-identified sentiment analysis, knowledge extraction, chatbots, and recommendation systems. A similar overlap was observed between the “autonomous systems & robotics” use case and the “automation” subfield. The following breakdown considers AI-related terms, and each subsection describes the use of these terms in specific phases related to product management. Table 3 summarizes the applications of AI found above for each phase of product development and management.

4.1 Demand forecasting

In the literature, demand forecasting is closely related to Supply Chain Management, e.g., in the electronics, healthcare, or hotel markets (Dombi et al. 2018; HajMirzaei et al. 2020; Li et al. 2021a). As written in van Steenberg & Mes (2020): “Accurately forecasting market demand of a product is essential for its design, distribution, promotion, and pricing strategies”. Therefore, this method will be most relevant in the phases related to the full commercialization of the product and its subsequent management. AI-enabled demand forecasting is described in industries such as, for example, fashion, food, or beauty (Ren et al. 2017, 2020; Gružasuskas et al. 2019; Jin

Table 3 AI methods and technologies used in each phase of product development and management, with descriptions

Phase	Method/technology	Description/aim
Idea generation	Sentiment analysis Knowledge extraction Conversational AI Recommendation systems Other ML algorithms	Assessing the positivity and negativity of product reviews Extracting product ideas from online sources Chatbots for customer interviews, knowledge-based systems for creativity improvement Information overload reduction Extracting product ideas from online sources
Concept testing and evaluation	Knowledge extraction	Concept evaluation within the organization
Product design and development	Sentiment analysis Knowledge extraction Conversational AI Computer vision Autonomous systems and robotics Other ML algorithms	Evaluation of how the product design was received Extracting product ideas from online sources Expert systems as product configurators for better product customization Help in the product manufacturing processes Help in the product manufacturing processes Researching design ideas from online sources
Product testing and validation	Computer Vision Anomaly detection Other ML algorithms	Product quality management Product quality management Product quality and testing
Product launch	Demand forecasting	Proper distribution of products to stationary stores
Product improvement and optimization	Sentiment analysis Conversational AI	Evaluation of how the product features were perceived Expert systems for quality enhancement
Product management	Demand forecasting Knowledge extraction Conversational AI Computer vision Anomaly detection Recommendation systems Other ML algorithms	New product forecasting, demand forecasting for a remanufactured product or spare parts Competitor analysis, customer churn Chatbots for product sales calls Social media image analysis and video mining for product placement and customer profiling Social media customer profiling Recommending products in the sales process NPD strategy suggestion, prescriptive analytics, store layout and location, product price optimization, churn prediction, personalization, choice modeling

and Shin 2020; Souza et al. 2020; Tsao et al. 2023) and can help to properly distribute products to stationary stores, resulting in no overstocking, higher customer satisfaction and retention. Moreover, demand forecasting can also be used to forecast the demand for spare parts for regularly sold products (Gonçalves et al. 2021).

Some studies (van Steenberg and Mes 2020; Cao and Zhang 2021; Kharfan et al. 2021; Li et al. 2021b) indicate a separation between demand forecasting, which deals with existing products in the market, and new product forecasting, for which there is no historical data to help determine future demand. To cope with such a task, various AI-based methods are given, e.g., based on historical data of similar products to those to be introduced for the first time (Lee et al. 2014; van Steenberg and Mes 2020; Kumar and Nagpal 2024; Saunders et al. 2024).

Some studies also mention demand forecasting for remanufactured products (Tsiliyannis 2018; Van Nguyen et al. 2020). Remanufacturing itself has grown into a significant industry because of both economic and environmental benefits. However, there is a great deal of uncertainty involved in determining demand due to the fact that it is unknown when and in what condition the product will return to the manufacturer, which in turn leads to a heavy estimation process of disassembling, inspecting, and testing the product.

4.2 Sentiment analysis

Another method used in the product management process is sentiment analysis (also referred to as opinion mining in the literature). In recent years, many articles have described the current state of research on sentiment analysis, its algorithms, and its uses in different domains (Mukhopadhyay 2018; Choi et al. 2020b; Birjali et al. 2021; Verma and Yadav 2021). Sentiment analysis to study needs data sources, such as social media, review websites, blogs, forums, or interview transcripts. It uses various approaches to extract this data from them, including the machine learning approach. The method is being developed so that it does not only distinguish between positive and negative comments but to introduce a fuzzy level of negativity and positivity, which can lead to better results (Aguwa et al. 2017; Giannakis et al. 2022; Li et al. 2023; Yang et al. 2024) or complement the sentiment information also with information about exactly which aspect the customer likes and has a negative attitude towards (Li et al. 2019; Ettrich et al. 2024; Kim and Cho 2024).

Sentiment analysis is frequently used in the product idea generation phase (Escoffier et al. 2018; Lee et al. 2018; Zhang et al. 2018; Mohammadian et al. 2025). Research confirms that it is used in both open innovation communities (i.e., internal company sources plus sometimes friendly rivals) and crowdsourcing (external sources, mainly customers) to explore others' opinions on a product idea. What is more, in searching for product ideas, the mentioned social media are often used, where sentiment analysis is combined with topic modeling (specifically, with the Latent Dirichlet Allocation method). Topic modeling is then responsible for determining how important a product topic is, and sentiment analysis is responsible for quantifying satisfaction with the product introduction (Jeong et al. 2019; Verza et al. 2023).

The literature also contains numerous studies on the impact of sentiment analysis on product design (Wang et al. 2018; Yagci and Das 2018; Yang et al. 2019; Akbari et al. 2020; Giannakis et al. 2022; Kim et al. 2023; Ramshankar and Joe Prathap 2023; Maarif et al. 2024; Ririn and Astanti 2024; Lai et al. 2024), product development (Choi et al. 2020a; Yoon et al. 2020; Purwandari et al. 2023; Anny Leema et al. 2024; Panigrahi et al. 2024), or product improvement (Yang et al. 2021; He 2024; Li et al. 2024b; Tripathy et al. 2024; Xu et al. 2024; You et al. 2024). Most of them mention that the increase in user-generated content (UGC) in social media influences designers to reach out to the opinions of people who are, or in the future may become, users of their products. On the other hand, it is worth noting that in this way, it is only possible to find out the opinions of users who are affected by sentiment analysis, but not how to spread the popularity of a product so that it becomes successful (Akbari et al. 2020). By studying UGC, it is also possible to learn the sentiment of users of one product and identify the advantages (or disadvantages) that a product has over its competitors (Liu et al. 2019; Wang et al. 2023c; Ma et al. 2024).

4.3 Knowledge extraction

Knowledge extraction (or information extraction) is another method belonging to NLP, strongly associated with sentiment analysis or topic modeling. In the scientific literature, it is usually accompanied by the terms "knowledge management", "text mining", or "data mining".

Knowledge extraction, like sentiment analysis, is used in the idea generation phase. Sources can still be online reviews (Luo et al. 2019b; Zhang et al. 2021b; Asadabadi et al. 2023; Güneş 2023; Sai et al. 2023; Li et al. 2024c; Lin et al. 2024; Seifian et al. 2024; Iuga and Brad 2025), patent information (Nikabadi and Sani-Varjovi 2021), crowdsourcing communities (Hwang et al. 2019; Imam et al. 2024) or external study participants (Aydogan et al. 2024), but the information that is extracted is more detailed because it is not only about user sentiment, but, for example, it is possible to retrieve innovative product ideas (Ghane and Cavallucci 2025). Moreover, knowledge extraction can also help evaluate ideas internally to ensure that potentially valuable opportunities are not missed (Yaakobi et al. 2019). In the later stages of product management, knowledge extraction can also be used for product design (Leung et al. 2010; Lai et al. 2019; Park et al. 2023; Shi et al. 2023; Zhang et al. 2024), competitor analysis (Guo et al. 2017), or customer churn (Jamjoom 2021).

4.4 Conversational AI

Research on conversational AI in product development and management in the academic literature has focused mainly on using chatbots. They are primarily used to evaluate customer experience and customer satisfaction (Sidaoui et al. 2020; Chen et al. 2021; Eren 2021; Balakrishnan and Dwivedi 2021; Baabdullah et al. 2022; Alabed et al. 2022) or directly in the sales process (Luo et al. 2019a). Combining

chatbots with sentiment analysis (Sidaoui et al. 2020) allows us to learn people's sentiments towards a product directly from many interviews without the involvement of a human role from the manufacturer's side. However, the use of chatbots comes with several challenges. X. Luo et al. (2019a, b) investigated that chatbots can be as good at the sales process as experienced employees and up to four times better than their inexperienced counterparts, but only if the chatbot did not reveal from the beginning that it was not human. Disclosing the chatbot at the beginning caused a drastic drop in sales due to the perception of chatbots as incompetent and indifferent. Moreover, it is sometimes difficult to assess the usability of a particular chatbot in a specific business (Borsci et al. 2022) due to the complexity of implementing a solution that has, for example, a tailored writing style or "character" to the audience.

When talking about conversational AI, one should also mention expert systems, which are most often used to help with product design and product development (Chakraborty and Boral 2017; Shafiee et al. 2021; Szejka et al. 2024) and with product improvement or quality management (Mouludi et al. 2019). One of the more exciting uses of expert systems is using them as the technology behind a product configurator, a tool to aid product design and customization (Shafiee et al. 2021). Conversational AI in the form of knowledge-based systems can also help in the idea generation phase to boost the creativity of teams by suggesting appropriate creativity and innovation techniques (CITs) to use with specific customer demands (Botega and da Silva 2020).

4.5 Image recognition, object identification, object tracking, and vision

Research related to computer vision issues is based mainly on product development and corresponding quality management (Jain et al. 2020; Liao et al. 2021; Zhao et al. 2022; Choi et al. 2024), which is in some cases related to the use of robots in the product development process, which will be written about in the next paragraph. Moreover, the research also mentions using computer vision solutions in design (Vijayaraj et al. 2023) or remanufacturing, for on-site part identification, to save time finding the correct spare parts for a product (Lehr et al. 2020).

In addition to examining product defects, computer vision solutions are also used to analyze human behavior (Rutter et al. 2021; Schwenzow et al. 2021; Micu et al. 2022). Social media image analysis (Rutter et al. 2021) and video mining (Schwenzow et al. 2021) are used to learn about consumer engagement with a product on platforms other than the manufacturer that released it. Such analyses can help in better product placement and appropriate advertising strategies. Other studies go a step further, and instead of social media platforms, they do customer profiling on-site, directly in the store (Micu et al. 2022) to further personalize products for the desired audience.

4.6 Autonomous systems and robotics plus automation

When describing the research related to autonomous systems and robotics, we combined it with the automation subfield because these concepts often occur together. It

is hard to find in the scientific literature the use of autonomous systems, robots, or simply automation in creating or managing a product, especially since robots themselves can be considered a product (Bertolini and Aiello 2018). The most frequent mentions refer to the use of robots as companions of humans in the product creation process or even instead of humans, performing simple automated tasks while humans can focus on more complex ones that require some decision making (Shukla et al. 2017; Dalmarco et al. 2019; Kose and Sakata 2019; Tubadji et al. 2021; Johnson et al. 2022; Pietronudo et al. 2022). Nevertheless, when combined with other methods from the field of NLP, CV or expert systems can achieve a high degree of autonomy and help in more demanding tasks than purely mechanical ones, such as planning or forecasting (Blöcher and Alt 2021). Some researchers point to the use of simulation environments consisting of software and potentially a physical robotic arm that can be used in the early stages of idea generation, prototyping and evaluation (Emektar et al. 2024; Truss and Schmitt 2024; Pascher et al. 2024). Correspondingly, some research considers software robots, as Robotic Process Automation (RPA) is called, as part of robotics (Shukla et al. 2017), due to the fact that it improves simple, routine tasks.

4.7 Anomaly detection

Anomaly detection is another method whose task is to analyze a given text, but this time to find anomalies in it. There are few applications of anomaly detection in the scientific literature that can be found in the process of creating or managing a product. A method classically used more for analyzing the logs of a given business process (Sarno et al. 2020; Ko and Comuzzi 2021), it can also be used for reviews or social media analysis (Mahmood and Alanezi 2021; Cui et al. 2024), e.g., to detect users from a given collection who do not fit the description of the target customer to whom the dedicated product would be delivered. Anomaly detection is also used in the industry in the product quality management process (Langone et al. 2020; Garg et al. 2021).

4.8 Recommendation systems

Information overload has led to the development of recommendation systems that suggest the most interesting information to the user, especially popular in the e-commerce industry. In our research, however, what is important is not how to implement a recommendation system but how such a system can help in the product management process.

In the academic literature, most approaches focus on which recommendation system is the best from a technical point of view (Cunha et al. 2018; Chen et al. 2019; Sánchez and Bellogín 2019), focusing on parameters such as accuracy, variety, or efficiency. However, there is a smaller amount of approaches in which researchers look at the solution from a business perspective, taking into account different business considerations, yet this directly affects how the company is perceived by users and thus directly affects product sales (Gorgoglione et al. 2019). Even the

best-suited recommender systems may face challenges such as a decrease in sales diversity (compared to companies that do not use recommender systems at all) (Lee and Hosanagar 2019) or the so-called cold-start problem, where at the beginning of its life, a recommender system has no data from which to recommend something (Zhang et al. 2017) or at least the fact that users may simply be tired of recommender systems and therefore do not use them (Yin et al. 2018). A particular solution to the problems mentioned above may be the combination of social networks and recommendation systems (Deng et al. 2017), which may positively affect sales diversity and the cold-start period.

Recommendation systems can also be used in combination with other methods (Lee et al. 2018; Ai 2023; Wang et al. 2023b, 2024; Chen et al. 2024; Gao et al. 2024; Pujari et al. 2024; Saed et al. 2024; Yu and Li 2024), such as image detection or the previously mentioned sentiment analysis in the idea generation phase. Knowing user sentiment from hundreds of reviews is still much information to process, so a recommender system allows the user to select the most valuable ones, increasing the efficiency of the whole process.

4.9 Machine learning, AI optimization, and predictive analytics

Machine learning (ML), in general, as a technology, is ubiquitous in the product development and management process. In the description, due to the small number of articles on predictive analytics and AI optimization in the product management process and similar content, we combined this into one subsection.

In the idea generation phase for extracting ideas from crowdsourcing communities and other online communities (Christensen et al. 2017, 2018; Hoornaert et al. 2017), from patent information (Woo et al. 2019; Jang et al. 2023; Liu et al. 2023b; Li et al. 2024a), or analysts' reports (Bellstam et al. 2021) machine learning solutions are commonly used. These can also help to aid in brainstorming (Jiun Gan et al. 2023). In addition to the methods mentioned earlier, other linear and nonlinear algorithms (Hoornaert et al. 2017; Joung and Kim 2023; Sanshi et al. 2023) related to text mining in the broadest sense are also used here. ML-based solutions assist also in various cases with product design and development. For example, convolutional networks are used to check user needs with user-generated content (Timoshenko and Hauser 2019). ML solutions are helping predicting the aesthetic scores (Burnap et al. 2023), predicting sustainable patterns in consumer product preferences (Chen 2023), optimizing design parameters (Lackes and Sengewald 2023; Raoalathi et al. 2023; Wang 2023; Wang et al. 2023a; Zhang et al. 2023), or reducing the dimensions, limiting possible design evaluations (Song and Liu 2023). Even ML-based customer grouping for product design is used (Suguna et al. 2019). Moreover, these solutions support physical (Wu et al. 2022; Deb et al. 2024; Lobo et al. 2024; Shi et al. 2024; Silva et al. 2024) or software (Naresh et al. 2019; Goyal 2020; Singh et al. 2023) product quality and testing and even help in choosing the right strategy for whole new product development (Wang et al. 2020).

Articles on predictive analytics in product management focus even more on the issue of prescriptive analytics (Viswanandhne et al. 2019), i.e., not what will happen

in the future, but what action must be taken now to eliminate the problem in the future. This is important, especially when it comes to portfolio management, cost optimization, product returns or customer-targeted product campaigns (Holmes and Kumar 2023; Xu et al. 2023; Hammann 2024; Sleiman et al. 2024; Choudhury et al. 2024; Duong et al. 2025). Artificial intelligence mechanisms also support automating and optimizing processes that happen not directly but around product creation, e.g., when dealing with sub-suppliers (Allal-Chérif et al. 2021). Already after the product creation process, during the development of the product life cycle, there are many activities related to product management, in which machine learning solutions help in. One of them is, for example, clustering for store assortment, layout or even for a store location to maximize product sales (Huang et al. 2019; Lee and Shin 2020; Yu et al. 2024). Others are applications related to risk optimization (Liu et al. 2024), product price optimization or price prediction (Ferreira et al. 2018; Simchi-Levi and Wu 2018; Greenstein-Messica and Rokach 2020; Zhu et al. 2024), e.g., dynamic pricing algorithm based on Thompson sampling (Ferreira et al. 2018), but also what effect product pricing has on customer behavior (Volkenfelt and Situmeang 2020). Other research related to customer behavior is related, for example, to how a customer will behave in a store after seeing a product online (Dzyabura et al. 2019) or churn prediction (Pustokhina et al. 2021). Also, reinforcement learning solutions find their applications, e.g., for product concept development and evaluation (Fazeli and Peng 2024), dynamic pricing challenges (Liu et al. 2023a), learning from competitive experience (Luoma et al. 2017) or the challenges of personalizing services to each customer (Byrd and Darrow 2021). Related to the above are interesting studies focused on choice modeling from the social sciences, related to causal analysis (Liebe and Meyerhoff 2021).

4.10 Experts' responses

The expert consultations provided a multifaceted perspective on both the current state and future directions of AI in product development and management, while also shedding light on the major AI applications currently deployed in these areas. Among these, AI-based forecasting and demand prediction were frequently mentioned as essential for optimizing inventory and supply chain processes. Similarly, sentiment analysis and natural language processing tools—often employed through chatbots or knowledge extraction systems—are widely used to gauge customer feedback and refine product concepts. One expert explained, “Tools that analyze online reviews and social media data are crucial for understanding customer sentiment, which in turn informs product design and iterative development. But it’s not used in every case.” Additionally, the integration of computer vision in quality control and autonomous systems in manufacturing processes was highlighted as a critical area where AI is already making significant inroads.

Beyond these applications, experts emphasized that AI’s value extends beyond isolated tools. They stressed that the real breakthrough comes from integrating these applications across the entire product lifecycle. “AI is not just an IT or data science issue; product managers, marketers, and even frontline employees need to

be involved. The biggest breakthroughs come when teams pool their expertise to refine AI models and interpret outputs,” remarked one expert. This interdisciplinary approach is essential for transforming point solutions—such as forecasting or sentiment analysis—into cohesive, end-to-end systems where insights seamlessly inform every phase, from ideation and concept testing to post-launch analysis.

One recurrent theme was the importance of evaluating an organization’s maturity and readiness before launching any AI initiatives. As one expert noted, “Before any AI pilot, organizations need to evaluate whether they’re truly ready—technologically and culturally. I’ve seen many AI projects fail simply because the company didn’t have the right data infrastructure or a supportive leadership.” This observation underscores the necessity for robust data governance. As one expert succinctly put it, “People underestimate how critical data governance is. Garbage in, garbage out. If you’re not systematically ensuring data accuracy, your AI won’t produce actionable insights.”

Looking to the future, experts identified several promising avenues for further research and development. They called for the creation of adaptive AI systems that continuously learn from new data and real-time feedback, ensuring that the technology remains relevant throughout a product’s lifecycle. The development of scalable frameworks and standardized guidelines was also seen as essential, particularly for smaller organizations that may lack the resources to develop custom solutions. Moreover, the need for human-centered design in AI tools was highlighted, with experts stressing that understanding the psychology of user interactions with AI is just as important as technical performance. “It’s not just about accuracy—it’s about how comfortable people feel delegating certain decisions to an algorithm,” one expert observed. Finally, the potential impact of generative AI on product innovation was noted as a transformative yet underexplored area. While generative AI could radically change how new products are conceptualized and prototyped, further empirical studies are needed to determine whether these technologies will spur disruptive innovation or merely accelerate incremental improvements.

5 Discussion

5.1 RQ1. What are the current major themes and research gaps in the literature when it comes to AI that helps with product development and management processes?

The existing academic literature on Artificial Intelligence (AI) in product development and management demonstrates considerable progress, yet it also reveals certain limitations and underlying assumptions that merit further investigation. Several challenges arose during the search for relevant solutions and their corresponding descriptions. Firstly, the cross-sectional nature of AI methodologies often led to ambiguities in phase-specific searches. For example, when searching for articles pertaining to the product design phase, results frequently referred to unrelated phases, such as idea generation, or vice versa, due to the overlapping application of certain AI techniques across multiple phases. Secondly, despite attempts to refine the

search terms, a significant number of articles appeared under categories that ostensibly related to business analysis but were unrelated to the actual product development and management processes. This overlap was a consequence of the sometimes overly broad terminology used, which resulted in the inclusion of works from non-relevant fields. Thirdly, a notable portion of the articles retrieved focused on the management of AI-enabled products, as opposed to an exploration of how AI can specifically assist in the product development and management processes.

The expert interviews generally reinforced the literature-based findings, with several experts emphasizing that AI's value lies in its potential to enhance decision-making and streamline workflows across the product lifecycle—from ideation to post-launch optimization. However, some experts also pointed out the challenges of integrating AI in later stages of product management, where organizational inertia and legacy systems can hinder adoption.

Upon examination of the findings and Table 3, several areas of substantial research growth alongside potential research gaps, as outlined in Table 4, can be identified. These gaps can be attributed to a variety of factors. First, AI solutions may prove ineffective in certain phases, rendering their attempted application suboptimal. Second, some specific AI methods—such as demand forecasting—may not be appropriate during certain stages, such as idea generation, where investing too many resources in preliminary concepts may not yield fruitful results. Third, employing an alternative breakdown of the product management phases may facilitate the discovery of additional relevant articles that were otherwise missed by the current keyword-based approach. Additionally, cross-phase combinations—such as integrating conversational AI into the concept testing and evaluation phase—deserve scholarly attention. For instance, in concept development, an AI-based system could pose questions to assess whether a concept aligns with business assumptions or even suggest areas for testing based on prior evaluations within the organization.

While AI applications such as knowledge extraction, sentiment analysis, and recommendation systems have been widely adopted, the focus has been disproportionately skewed toward early-stage activities such as idea generation and design support. A major issue with the existing research is the fragmented application of AI methods across different phases of product management, which results in a lack of continuity and strategic integration. AI-driven insights generated in one phase often remain siloed, failing to influence decision-making in subsequent stages. This problem stems from the assumption that AI can be deployed in isolated use cases rather than as a continuous and evolving intelligence layer spanning the entire product lifecycle. Future research must explore how AI can function as an integrative force, ensuring knowledge transfer and iterative learning across all phases of product development.

Another overlooked challenge is the implicit assumption that AI implementation in product management is universally beneficial. While AI has demonstrated potential in increasing efficiency and optimizing decision-making, few studies critically examine the unintended consequences of AI-driven decision systems. For example, reliance on AI-driven product ideation risks prioritizing incremental improvements over radical innovation, as AI models are trained on existing datasets rather than envisioning entirely new possibilities. Additionally, current literature assumes

a static relationship between AI and product development, treating AI as a tool to be implemented rather than a force that continuously reshapes market landscapes. However, AI's potential to dynamically redefine consumer preferences or shift competitive advantages is rarely discussed. Surprisingly, the search results lack articles describing AI-based experiment design in the product development and management process. Searching by selected keywords, there are also no descriptions of methods such as AI-based A/B/n testing, which is in some cases an essential step in the process, resulting in another potential research gap.

5.2 RQ2. What would it take to create a common AI-driven product development and management framework?

The analysis presented in Tables 3 and 4 highlights a wide range of AI solutions applicable across various stages of the product development and management lifecycle. To systematize the existing literature and enhance the utility for both researchers and practitioners, it is imperative to develop a unified AI-driven framework for product development and management. Such a framework would aid in identifying research gaps, streamlining processes, and providing a coherent understanding of how individual studies fit within the broader research context.

Creating a comprehensive framework involves several critical considerations. These considerations were previously addressed in this paper and include, for example, the nature of the product—whether physical, software, or a hybrid (Porter and Heppelmann 2015)—and the size and industry of the organization are essential criteria (Schuh et al. 2019; Borah et al. 2022; Oldemeyer et al. 2024; Tominc et al. 2024). These factors influence the specific requirements and constraints of the product development and management process. For example, the processes in a startup differ significantly from those in a large corporation, and variations are also evident across industries such as agriculture, manufacturing, and medicine etc. (Vaia et al. 2021; Chatterjee et al. 2022; Taneja et al. 2023). To address these variations, frameworks can be developed using either perpendicular or parallel mapping approaches. Perpendicular mapping involves integrating AI across the entire organizational structure, ensuring a holistic adoption. In contrast, parallel mapping involves adapting a similar process or departmental framework, such as those used in innovation management, to the product development and management context (Füller et al. 2022; Bahoo et al. 2023).

Implementing AI solutions in product development and management necessitates meticulous planning and strategic alignment with organizational goals. Organizations need to define clear objectives for AI integration (why to use it), how these objective can be achieved (what methods to use), identify metrics and key performance indicators (KPIs) to evaluate success (how to measure it) (McKinsey 2021), ensure the availability of necessary resources and infrastructure (what challenges we need to overtake) (Woschke and Haase 2016; Figaliet al. 2020; Erebak and Turgut 2021; Omrani et al. 2022) and make sure it is aligned with their AI strategy (where are we/how mature are we on the AI implementation way) (Molla et al. 2009; Lokuge et al. 2019; Jöhnk et al. 2021). Exemplary guidelines and critical questions,

Table 4 Potential research gaps in product development and management process (PRG, potential research gaps, SRC, some research conducted)

	Idea generation	Concept testing & evaluation	Product testing & development	Product testing & validation	Product launch	Product improvement & optimization	Product management
Demand forecasting	PRG	PRG	PRG	PRG	SRC	PRG	SRC
Sentiment analysis	SRC	PRG	SRC	PRG	PRG	SRC	PRG
Knowledge extraction	SRC	SRC	SRC	PRG	PRG	PRG	SRC
Conversational AI	SRC	PRG	SRC	PRG	PRG	SRC	SRC
Vision*	PRG	PRG	SRC	SRC	PRG	PRG	SRC
Autonomous systems**	PRG	PRG	SRC	PRG	PRG	PRG	PRG
Anomaly detection	PRG	PRG	PRG	SRC	PRG	PRG	SRC
Recommendation systems	SRC	PRG	PRG	PRG	PRG	PRG	SRC
Other ML algorithms***	SRC	PRG	SRC	SRC	PRG	PRG	SRC

*Vision, Computer Vision, image recognition, object identification, object tracking

**Autonomous systems, Autonomous systems and robotics, automation

***Other ML algorithms, Machine Learning, AI optimization and predictive analytics

as summarized in Table 5, can assist in this endeavor. However, these guidelines are not exhaustive and can be tailored to fit specific organizational needs and combined to support the effective implementation of AI solutions.

5.3 Implications for theory and practice

A recurring theme in the literature is the fragmented way AI tools are adopted in different phases of product management, often without a unifying conceptual framework. Scholars tend to examine AI applications in isolated, phase-specific silos—sentiment analysis in early ideation, demand forecasting in later commercialization, or computer vision in quality assurance—yet few studies link these discrete interventions into a continuous, learning-driven process. The resulting fragmentation suggests that current theories of product development do not fully account for how AI might operate as a cross-cutting, integrative force throughout the entire product lifecycle. This gap opens space for more holistic theoretical models that view AI not simply as a standalone tool, but as an evolving layer of intelligence that can dynamically reshape market landscapes, consumer preferences, and competitive dynamics.

Another theoretical tension arises from the assumption that AI necessarily drives more innovative or efficient product outcomes. While AI-based analytics can indeed reveal valuable patterns or automate routine tasks, it can also inadvertently privilege incremental improvements over radical innovations by relying on historical data. Such data-driven biases may perpetuate existing market structures and customer preferences, leaving open the question of how truly novel product concepts emerge. Future theories should explore how organizations can cultivate a more exploratory or generative AI use, one that balances data-driven insights with the unpredictability and risk-taking inherent in radical innovation.

One significant area with implications for both theory and practice are the development of common frameworks for AI integration in product development and management. Such frameworks should account for the specific needs and contexts of various industries, product types, and organizational sizes. By building on the guidelines outlined in this study—covering aspects such as methods used, reasons to use these methods, key performance indicators, and AI maturity—these frameworks can help standardize the application of AI, making it more accessible for organizations to adopt and benefit from AI technologies. This approach ensures that critical considerations—including data governance, risk management, and cross-functional collaboration—are systematically incorporated into AI-driven product management processes, effectively bridging the gap between theoretical insights and practical implementation.

From a managerial standpoint, the integration of AI into product management calls for strategic coherence and organizational readiness. While many firms experiment with AI tools—chatbots for user feedback, recommendation engines for product ideation—these experiments often remain isolated pilots. To realize AI's full potential, organizations need clear objectives regarding why and how AI solutions will be leveraged across different product phases, accompanied by a roadmap for scaling and sustaining these solutions. Establishing a robust data infrastructure and

ensuring adequate team capabilities—both technical and managerial—are essential first steps. In practice, this includes assembling cross-functional teams that blend product domain knowledge with AI expertise, thereby avoiding the siloed decision-making that can diminish AI's overall impact.

Lastly, the literature's focus on narrow, phase-specific metrics for AI success (e.g., accuracy in demand forecasting) may obscure broader measures of product performance and innovation. Managers can benefit from designing multi-faceted performance indicators—covering market adoption, cultural impact, or radical innovation outcomes—that capture AI's contributions across the product lifecycle. Such holistic measurement can reveal how AI's benefits in one stage (e.g., ideation) might reverberate into another (e.g., commercialization), helping organizations allocate resources more effectively and identify where AI-driven experimentation adds the most value.

5.4 Limitations and future directions

This study, while comprehensive in its approach, has several limitations that need to be acknowledged. Firstly, the categorization of AI methods and their applications in the product management process is somewhat subjective. Different researchers might categorize and interpret the AI methods and their applications differently, leading to variations in findings. The study also relies heavily on the definitions and terminology used in the literature, which can vary widely and sometimes lack

Table 5 Exemplary guidelines on building an AI-driven product development and management framework

Guidelines/questions	Examples
What is the type of product? (Porter and Heppelmann 2015)	Software/Physical (Hardware, ...)/Smart
What is the type/size of the organization? (Schuh et al. 2019; Borah et al. 2022; Oldemeyer et al. 2024; Tominc et al. 2024)	Start-ups/SME/Big companies/...
What is the described industry? (Vaia et al. 2021; Chatterjee et al. 2022; Taneja et al. 2023), <i>VOS Viever map from this study</i>)	Agriculture/Medicine/Economics/...
What method of building framework is going to be used? (Füller et al. 2022; Bahoo et al. 2023)	Mapping/Synthesis of research/...
Why to use AI method in particular stage? (<i>Based on this study</i>)	To extract product ideas from online sources quicker/To improve product quality/...
What methods can be used? (<i>Based on this study</i>)	Sentiment analysis/Conversational AI/...
How the success can be evaluated? (McKinsey 2021)	CSAT/NPS/ROI/...
What are the current challenges? (Woschke and Haase 2016; Figalist et al. 2020; Erebak and Turgut 2021; Omrani et al. 2022)	Data quality/AI Strategy/Data privacy/...
What is the AI readiness/maturity level? (Molla et al. 2009; Lokuge et al. 2019; Jöhnk et al. 2021)	AI-frontrunners/AI-practitioners/...

clarity. This can lead to potential misinterpretations and overlaps in categorizing AI methods and their applications. Additionally, the selection of keywords and databases, although carefully chosen, might not cover all possible articles related to the integration of AI in product development and management. This could result in a partial view of the current state of research in this area. Lastly, the research does not delve deeply into the economic, social, and ethical implications of AI integration in product development and management. These factors are crucial for a holistic understanding of AI's impact but were beyond the scope of this study. Future research should aim to address these aspects to provide a more rounded perspective.

Despite the limitations, this study opens several avenues for future research in the domain of AI-driven product development and management. One promising area for future research is the exploration of specific AI methods in under-researched phases of product development and management, as identified in this study. For example, further studies could investigate the use of conversational AI in concept testing and evaluation or the application of anomaly detection in product design and development. These explorations could uncover new insights and practical applications, enriching the existing body of knowledge. Building on the identified limitations and the research gaps outlined in Tables 3 and 4, future work could explore more integrative frameworks that transcend siloed, phase-specific AI applications. Researchers might investigate how AI can function as a continuous, adaptive layer of intelligence, informing decisions from early ideation through post-launch optimization. In the method descriptions, we pointed out that, for example, in one article, a chatbot does customer interviews. Then sentiment analysis is used to determine customer attitudes toward the product. In another article, sentiment analysis was done on online reviews, after which a recommendation system sorted through this data and made indications for the manufacturer. Such solutions could be combined into one, i.e., using chatbots for interviews, sentiment analysis, and recommendation system in one case. Such combinations of AI methods across multiple phases of product development and management process could be an interesting research gap for future researchers and information for managers on how to differentiate themselves in the market.

Furthermore, there is a need for more empirical studies and real-world case studies that demonstrate the successful implementation of AI in product development and management. Such studies would provide practical insights and examples, helping bridge the gap between theory and practice. They could also highlight best practices, common pitfalls, and effective strategies for overcoming the challenges associated with AI integration. There is also a compelling need for longitudinal, mixed-method studies that capture the evolving role of AI within product management teams. Quantitative analyses of performance metrics (e.g., time-to-market, cost savings, revenue growth) could be combined with qualitative insights (e.g., employee interviews, observational studies) to provide a richer understanding of how AI reshapes team dynamics, creativity, and strategic thinking over time. These investigations would be especially valuable in industries where AI adoption is nascent, helping to surface barriers to implementation and strategies for overcoming them. Another promising avenue lies in cross-disciplinary research, bringing together perspectives from management, computer

science, ethics, and behavioral psychology. Collaborations across these domains could lead to more robust AI solutions that incorporate explainability, fairness, and user-centric design principles—an increasingly critical concern as AI systems become more autonomous and complex.

Finally, future studies should consider the impact of generative AI—a rapidly advancing area that extends beyond predictive analytics and automation. Generative models, capable of producing novel product concepts or design variations, open new questions about the locus of innovation: Will AI primarily serve as a creative partner to human teams, or might it displace certain human roles altogether? Investigating how generative AI might accelerate or impede radical innovation could deepen theoretical debates on AI's long-term consequences for product development and management.

6 Conclusions

This study has explored the role of Artificial Intelligence (AI) in the product development and management process. Our analysis reveals that while there is a wide range of approaches to AI and product development or management in the literature, there is limited integration of these two fields. Upon decomposing these approaches into specific methods and phases, it becomes apparent that certain areas of research have been more thoroughly explored, while others have yet to be fully explored. This lack of cohesion in the literature highlights the need for the guidelines—the proposed guidelines should facilitate the systematic positioning of the AI-driven product development and management research in the scientific literature. The integration of AI, particularly in the context of New Product Development (NPD), offers considerable potential for enhancing various stages of product management, from idea generation to commercialization and optimization. The findings from the literature review and expert insights indicate that AI can provide substantial support in automating product management tasks, ultimately leading to more efficient and innovative outcomes.

In conclusion, expanding the analytical lens beyond current phase-specific uses of AI, incorporating cross-disciplinary and longitudinal methods, and addressing emerging technologies such as generative AI will enrich our collective understanding of how AI can transform product development and management. These future directions hold the potential not only to refine theoretical frameworks but also to guide practitioners in leveraging AI more ethically, strategically, and effectively across the entire product lifecycle.

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Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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